Guide 06: The Hyperlinked Diagnostic Retrieval Engine ("Do Now" 2.0)

Closing the Loop: Spaced Retrieval Practice, Micro-Explanation Drawers & Multi-Window Narrative Deep-Links

Executive Summary: The History Revision Hub has upgraded its lesson starter architecture from a static recall list into an **active diagnostic feedback loop**. In traditional classrooms, starters suffer from the *"Stuck Starter" Dilemma*: when pupils miss a factual question, the teacher must either interrupt lesson momentum for 10 minutes of verbal exposition or leave pupils with uncorrected misconceptions. The new Hyperlinked Retrieval Engine solves this by pairing 10-question spaced retrieval with instantaneous micro-context drawers and deep hyperlinks targeting the exact 1-paragraph narrative exposition in a non-destructive second tab.

COGNITIVE SCIENCE & DISCIPLINARY FOUNDATIONS

🧠 ROSENSHINE'S PRINCIPLES (1 & 10)

Daily & Spaced Review: Regular retrieval of prior knowledge strengthens memory traces and automates recall of foundational schema (e.g. 1973 Yom Kippur War, 1956 Suez Crisis), freeing working memory for complex causal analysis.

🔄 BJORK'S DESIRABLE DIFFICULTIES

Spaced Interleaving: Blocking questions by single lessons creates illusions of mastery. Interleaving questions across prior units forces active discrimination between parallel historical mechanisms and treaty terms.

📖 CHRISTINE COUNSELL'S NARRATIVE

Contextualized Recall: Historical facts are not isolated flashcard trivia; they are causal dominoes. Linking retrieval questions back to the narrative paragraph re-anchors facts into their authentic historical storyline.

THE 3-MODE RETRIEVAL MATRIX

RETRIEVAL MODE	SAMPLING ARCHITECTURE & SCOPE	CLASSROOM PURPOSE & COGNITIVE GOAL
Mode 1: Last Lesson **Immediate Prior Recall**	10 questions sampled strictly from the immediately preceding lesson (Lesson *N-1*). Drawn from verified lesson quizzes and flashcards.	Consolidates yesterday's content, verifies homework recall, and primes schema before introducing new causal mechanisms.
Mode 2: Unit Interleave **Spiral Topic Recall**	10 questions interleaved proportionally across all prior lessons in the active unit (Lessons 1 to *N-1*).	Prevents early-unit knowledge decay and trains pupils to distinguish between related treaties, dates, and individuals.
Mode 3: Synoptic "All" **Multi-Unit Revision**	10 synoptic questions cross-sampled across selected GCSE specifications (Middle East, USA, Weimar, Medicine, Elizabeth, Cold War) with toggle chips.	Simulates real terminal examination conditions where pupils must pivot fluidly across disparate periods and themes.

STRICT PEDAGOGICAL QUALITY GATES & ZERO-ECHO INTEGRITY

Rule 1: Strict Zero Same-Lesson Recall: A Do Now must NEVER ask questions about the current lesson's unlearned content (e.g. asking about the Camp David Accords at the start of the Camp David lesson). Retrieval must strictly draw from prior lessons.
Rule 2: Programmatic Source Gate: Questions referencing Source A–E extracts, cartoons, or paintings are automatically excluded, as pupils do not have documentary stimuli visible during starters.
Rule 3: Non-Tautological Explanation Standard: Micro-drawers only render if authentic historical context is available. Tautological echo text ("The answer is X because X") is programmatically suppressed, presenting a clean badge instead.

TECHNICAL ERGONOMICS & CLASSROOM PROTOCOL

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Part II: Multi-Window Tab Architecture, URL Routing & Standard Operating Procedures

ERGONOMIC MULTI-WINDOW ARCHITECTURE (ZERO LOST POSITION)

⚠️ THE LEGACY IN-PAGE NAVIGATION TRAP

In early builds, clicking a jump link navigated within the active tab:

- **State Erasure:** The active starter was unloaded, wiping all revealed answers and progress.
- **Timer Destruction:** Active 3-minute or 5-minute countdown timers were killed mid-starter.
- **Scroll Disorientation:** Returning required hitting 'Back', re-rendering the unit, and re-scrolling to the starter.

🛡️ THE NATIVE MULTI-WINDOW SOLUTION

All jump pills and drawer links enforce semantic `<a target="_blank" rel="noopener noreferrer">`:

- **100% Frozen State:** The primary Do Now starter remains completely undisturbed in the main tab.
- **Side-by-Side Reading:** The target lesson opens in a secondary tab, auto-scrolled to the exact stage.
- **Instant Dismissal:** Pupils read the 1-paragraph context, tap `Ctrl+W` to close, and continue the starter.

DEEP URL ROUTING & DYNAMIC IN-MEMORY DOM SPOTLIGHT

```
// Standard Deep-Link URL Architecture:
https://the-history-revision-hub.netlify.app/?view=lessons&unit=cme_new&lesson=9&stage=4
• unit: Target curriculum ID (e.g. cme_new, usa, weimar_nazi_germany, edexcel_medicine)
• lesson: 1-based sequential lesson index resolved via resolveLessonTarget(/^d+$/)
• stage: Exact narrative stage container (#para-4) centered and spotlighted with yellow glow
```

The Automated In-Memory Spotlight: Upon target page load, `src/engine/lesson_renderer.js` reads the `stage` parameter, queries `#para-[stage]`, executes `scrollIntoView({ behavior: 'smooth', block: 'center' })`, and applies the `.keyword-pulse-glow` animation (high-contrast yellow fill `#fef9c3` + 4px sky-blue focus outline `#0284c7`) for 3.5 seconds.

CLASSROOM STANDARD OPERATING PROCEDURES (SOP)

📽️ WHOLE-CLASS SMARTBOARD

Front-of-Class Delivery:

- Project starter; launch 5m countdown timer with chime.
- Pupils complete 10 items in exercise books or whiteboards.
- Tap rows to reveal model answers.
- Expand explanation drawer to clarify whole-class misconceptions on the fly.

💻 1:1 DIGITAL / INDEPENDENT

Self-Paced Retrieval:

- Pupil completes questions individually on Chromebook/iPad.
- On missed items, tap [📖 `Lesson X • Stage Y ↗`].
- Read the 1-paragraph narrative in a new tab (30-second context re-anchor).
- Close tab (`Ctrl+W`) with zero starter disruption.

🎯 DIFFERENTIATED INTERVENTIONS

SEND & Stretch Protocols:

- **Lower Attainers:** Clean, uncluttered answer pills provide immediate recall without cognitive overload.
- **Higher Attainers:** Encouraged to open narrative links to re-read surrounding causal mechanisms and historiography.